

COMPUTER NETWORKS

Part 1

Complete Study Notes

Unit 1 Introduction and Network Models

Unit 2 Physical and Data Link Layers

Part 1 of 3

UNIT 1 — Introduction and Network Models

1.1 What is Data Communication?

Data communication is the exchange of data between two devices through some form of transmission medium such as a wire cable or wireless link.

For data communication to be effective, it must satisfy four fundamental characteristics:

Characteristic	Meaning
Delivery	Data must reach the correct, intended destination — and only that destination
Accuracy	Data must arrive without being altered or corrupted in transit
Timeliness	Data must arrive in time; for audio/video it must arrive in the same order it was produced, without significant delay (real-time transmission)
Jitter	Variation in packet arrival time must be minimal. Uneven delays ruin the quality of video and audio

Five Components of Data Communication

1. **Message** — the information (data) to be communicated: text, numbers, images, audio, video.
2. **Sender** — the device that sends the message (computer, phone, camera).
3. **Receiver** — the device that receives the message.
4. **Transmission Medium** — the physical path the message travels through (cable, fiber, radio waves).
5. **Protocol** — a set of rules governing the communication. Without a shared protocol, two devices may be connected but cannot understand each other — like a person speaking Japanese to someone who only knows French.

1.2 Data Flow (Transmission Modes)

a) Simplex

- Communication is **one-directional only**.
- One device only transmits, the other only receives.
- The entire channel capacity is used in one direction.
- **Examples:** Keyboard → computer, computer → monitor, TV broadcast.

b) Half-Duplex

- Both devices can transmit and receive, but **not at the same time**.
- When one sends, the other must listen. The whole channel capacity is taken over by whichever device is transmitting.
- **Examples:** Walkie-talkie, CB radio.

c) Full-Duplex (Duplex)

- Both devices can transmit and receive **simultaneously**.
- The channel capacity is shared between the two directions, or two separate paths are used.
- **Examples:** Telephone call, mobile phone conversation.

1.3 What is a Network?

A **network** is a set of devices (often called **nodes** or **hosts**) connected by communication links. A node can be a computer, printer, server, phone, or any device capable of sending/receiving data.

Criteria a Network Must Meet

Criterion	Explanation
Performance	Measured by transit time, response time, throughput, and delay. Depends on number of users, type of medium, hardware capability, and software efficiency
Reliability	Measured by frequency of failure, recovery time after failure, and robustness in a catastrophe
Security	Protecting data from unauthorized access, damage, and implementing policies for recovery from breaches

Types of Connections

1. Point-to-Point — A dedicated link between exactly two devices. The entire capacity of the link is reserved for those two devices. *Example: TV remote and TV via infrared.*

2. Multipoint (Multidrop) — More than two devices share a single link. Capacity is shared either **spatially** (all share simultaneously) or **temporally** (devices take turns — time-shared).

1.4 Network Topologies

Topology is the geometric representation of how devices (links and nodes) are arranged relative to each other.

a) Mesh Topology

Every device has a dedicated point-to-point link to every other device.

- Number of links needed for n devices = $n(n-1)/2$
- Each device needs $n-1$ I/O ports

Advantages	Disadvantages
No traffic problems (dedicated links)	Huge amount of cabling and I/O ports
Robust — one link failure doesn't stop the system	Installation and reconnection is difficult
Privacy and security (dedicated path)	Hardware cost is very high
Easy fault identification and isolation	Space required for wiring is large

b) Star Topology

Each device has a dedicated point-to-point link only to a **central controller (hub/switch)**. Devices are not directly linked to each other. The hub acts as an exchange.

Advantages	Disadvantages
Less expensive than mesh	Single point of failure — if the hub fails, the whole network is dead
Easy to install and reconfigure	More cabling than bus/ring topology
Robust — one link failure affects only that device	
Easy fault identification	

c) Bus Topology

A **multipoint** topology. One long backbone cable links all the devices. Devices connect via **drop lines** and **taps**.

Advantages	Disadvantages
Easy installation, uses least cabling	Difficult reconnection and fault isolation
Ideal for small networks	Signal reflection at taps degrades quality
	A break in the backbone stops all transmission
	Adding new devices is difficult

d) Ring Topology

Each device has a dedicated point-to-point connection only with the two devices on either side. A signal passes around the ring in one direction, device to device, until it reaches its destination. Each device has a **repeater**.

Advantages	Disadvantages
Easy to install and reconfigure	Unidirectional traffic
Fault isolation is simple (alarm on signal failure)	A single break in the ring can disable the entire network (solved by a dual ring)
	Signal must pass through every node — adds delay

e) Hybrid Topology

A combination of two or more topologies. *Example:* A main star topology where each branch connects several stations in a bus topology. Most real-world networks are hybrid.

1.5 Types of Networks (by Geographic Scale)

PAN — Personal Area Network

- Range of about 10 meters, organized around one person.
- Technologies: Bluetooth, Infrared, Zigbee, NFC.
- Example: Smartphone connected to earbuds, smartwatch, and laptop.

LAN — Local Area Network

- Covers a single building, office, lab, or campus.

- Privately owned, high data rates (100 Mbps to 10 Gbps), very low error rate.
- Usually one transmission medium type; commonly uses Ethernet or Wi-Fi.
- Topologies: Star (most common today), bus, ring.

MAN — Metropolitan Area Network

- Covers a town or city (roughly 5–50 km).
- May be owned by a private company or a public utility.
- Examples: Cable TV network, a bank's branch network across a city, DSL line networks.

WAN — Wide Area Network

- Spans large geographic areas — countries, continents, or the whole world.
- Slower data rates than LANs, higher error rates, and usually leased from telecom providers.
- Two forms:
 - **Point-to-point WAN** — a single line leased between two points.
 - **Switched WAN** — a network with multiple endpoints, e.g., the backbone of the Internet.

Other Terms

- **WLAN** — Wireless LAN (Wi-Fi based).
- **CAN** — Campus Area Network (a group of interconnected LANs on a campus).
- **SAN** — Storage Area Network (high-speed network of storage devices).
- **Internetwork (Internet)** — two or more networks connected together, forming the global Internet.

Comparison Table

Feature	PAN	LAN	MAN	WAN
Range	~10 m	Up to a few km	5–50 km	Unlimited
Ownership	Private	Private	Private/Public	Usually public/leased
Speed	Low	Very high	High	Low to moderate
Cost	Very low	Low	Moderate	High
Error rate	Low	Lowest	Moderate	Highest
Example	Bluetooth	Office network	City cable network	The Internet

1.6 Switching

In a large network with many connected devices, we need **switching** — connecting devices to create temporary paths.

a) Circuit Switching

- A **dedicated physical path** is established between sender and receiver for the entire duration of the session.
- Three phases: **Setup → Data Transfer → Teardown**.
- Resources are reserved even if no data flows (wasteful).
- **Example:** Traditional telephone network (PSTN).
- *Advantage:* Guaranteed bandwidth, constant delay, no ordering issues.
- *Disadvantage:* Inefficient use of bandwidth, long setup time.

b) Packet Switching

Data is broken into **packets**, each handled independently.

Datagram approach (connectionless): - Each packet is treated independently and may take a different route. - Packets may arrive **out of order, duplicated, or lost**. - Used in the **Internet (IP layer)**.

Virtual Circuit approach (connection-oriented): - A logical (not physical) path is set up before transfer. - All packets follow the same route and arrive in order. - Uses a **Virtual Circuit Identifier (VCI)** instead of full addresses. - Example: Frame Relay, ATM.

c) Message Switching (Store-and-Forward)

- The whole message is sent to an intermediate node, stored completely, then forwarded.
- No dedicated path. Obsolete today but conceptually is the ancestor of packet switching.
- Example: Older telegraph/email relay systems.

Feature	Circuit Switching	Packet Switching
Path	Dedicated physical	Shared, dynamic
Setup required	Yes	No (datagram)
Bandwidth use	Reserved, wasteful	Efficient, on-demand
Delay	Fixed	Variable
Failure impact	Connection drops	Packets reroute
Suitable for	Voice	Data / Internet

1.7 Protocols and Standards

Protocol

A **protocol** is a set of rules governing data communication. It defines *what* is communicated, *how* it is communicated, and *when*.

Key elements of a protocol: 1. **Syntax** — the structure or format of data (e.g., first 8 bits = sender address). 2. **Semantics** — the meaning of each section of bits; what action to take. 3. **Timing** — when data should be sent and how fast it can be sent.

Standards

Standards ensure interoperability between products from different vendors.

- **De facto standards** — standards that became widespread through use, not by an official body (e.g., early Ethernet).
- **De jure standards** — formally legislated/approved by an official standards body.

Important Standards Organizations: - **ISO** — International Organization for Standardization (created the OSI model) - **ITU-T** — International Telecommunication Union, Telecommunication Sector - **ANSI** — American National Standards Institute - **IEEE** — Institute of Electrical and Electronics Engineers (802 series: 802.3 Ethernet, 802.11 Wi-Fi) - **EIA** — Electronic Industries Alliance (RS-232) - **IETF** — Internet Engineering Task Force (publishes **RFCs** — Request For Comments)

1.8 Layered Architecture — Why Layers?

Network communication is complex. Layering breaks it into smaller, manageable pieces.

Benefits of layering: - Each layer does a specific, well-defined job. - A layer can be modified or replaced without affecting other layers. - Simplifies design, troubleshooting, and standardization. - Allows products from different vendors to interoperate.

Each layer provides services to the layer above it and uses services of the layer below it. Communication between corresponding layers on two machines is called **peer-to-peer communication** and is governed by an **interface** and a **protocol**.

1.9 The OSI Reference Model

The **Open Systems Interconnection (OSI)** model was developed by **ISO** in 1984. It is a **7-layer** conceptual framework describing how data moves across a network. It is a *model*, not a protocol — it is not directly implemented.

The 7 Layers (Top to Bottom)

Layer 7 — Application Layer - The layer closest to the user; provides the interface for applications to access the network. - Services: file transfer, email, remote login, directory services, web browsing. - Protocols: HTTP, FTP, SMTP, DNS, Telnet, SNMP.

Layer 6 — Presentation Layer - Concerned with the **syntax and semantics** of exchanged information. - Functions: - **Translation** — converts between different data formats (ASCII ↔ EBCDIC). - **Encryption / Decryption** — for privacy. - **Compression** — reduces number of bits transmitted (important for multimedia).

Layer 5 — Session Layer - The **network dialog controller**. Establishes, maintains, and synchronizes sessions. - Functions: - **Dialog control** — allows half-duplex or full-duplex communication. - **Synchronization** — adds checkpoints into the data stream so that on failure, retransmission resumes from the last checkpoint rather than the beginning.

Layer 4 — Transport Layer - Responsible for **process-to-process delivery** of the entire message. - Functions: - **Service-point (port) addressing** — delivers to the correct process, not just the correct computer. - **Segmentation and reassembly** — divides messages into segments with sequence numbers. - **Connection control** — connectionless (UDP) or connection-oriented (TCP). - **Flow control** — end-to-end. - **Error control** — end-to-end; ensures whole message arrives error-free. - Data unit: **Segment** (TCP) / **Datagram** (UDP).

Layer 3 — Network Layer - Responsible for **source-to-destination (host-to-host) delivery of packets, possibly across multiple networks**. - Functions: - **Logical addressing** — IP addressing. - **Routing** — choosing the best path through interconnected networks. - Fragmentation, congestion control. - Devices: **Router**. Data unit: **Packet / Datagram**.

Layer 2 — Data Link Layer - Responsible for **hop-to-hop (node-to-node) delivery**; makes the physical layer appear error-free. - Functions: - **Framing** — divides the bit stream into frames. - **Physical addressing** — adds MAC addresses of sender and receiver. - **Flow control** — prevents overwhelming a slow receiver. - **Error control** — detection and

retransmission (trailer/CRC). - **Access control** — decides which device controls the link at a given time. - Devices: **Switch, Bridge, NIC**. Data unit: **Frame**.

Layer 1 — Physical Layer - Transmits the **raw bit stream** over the physical medium. - Concerned with: - Physical characteristics of interfaces and media. - Representation of bits (encoding into signals). - Data rate (bits per second), bit synchronization. - Line configuration (point-to-point / multipoint). - Physical topology, transmission mode (simplex/duplex). - Devices: **Hub, Repeater, Cables, Connectors**. Data unit: **Bit**.

Memory Aids

- Top → Bottom: "**All People Seem To Need Data Processing**"
- Bottom → Top: "**Please Do Not Throw Sausage Pizza Away**"

Grouping of Layers

- **Layers 1-3 (Physical, Data Link, Network)** — Network Support Layers
- **Layer 4 (Transport)** — links the two groups
- **Layers 5-7 (Session, Presentation, Application)** — User Support Layers

1.10 The TCP/IP Protocol Suite

The **TCP/IP model** was developed before OSI and is the model actually used on the Internet. It originally had **4 layers**; many texts describe it as **5 layers** by splitting Host-to-Network into Physical and Data Link.

TCP/IP Layers

TCP/IP Layer	Corresponding OSI Layers	Key Protocols
Application	Application + Presentation + Session	HTTP, HTTPS, FTP, SMTP, DNS, DHCP, SNMP, Telnet
Transport	Transport	TCP, UDP, SCTP
Internet (Network)	Network	IP, ICMP, IGMP, ARP, RARP
Network Access / Link	Data Link + Physical	Ethernet, Wi-Fi, PPP, Token Ring

OSI vs TCP/IP — Comparison

Aspect	OSI Model	TCP/IP Model
Number of layers	7	4 (or 5)
Developed by	ISO	US DoD / ARPANET
Nature	Theoretical reference model	Practical, implemented model
Protocol dependence	Protocol-independent (generic)	Built around specific protocols
Layer separation	Clear separation of service, interface, protocol	Less clearly separated
Session & Presentation	Separate layers	Merged into Application
Approach	Model first, then protocols	Protocols first, then model
Usage today	Teaching and reference	The actual Internet

1.11 Encapsulation and Data Units

As data travels **down** the layers on the sender's side, each layer adds its own **header** (and the data link layer also adds a **trailer**). This is called **encapsulation**. On the receiver's side, each layer removes its header — **decapsulation**.

Layer	Data Unit (PDU)	Added Information
Application	Data / Message	—
Transport	Segment (TCP) / Datagram (UDP)	Port numbers, sequence numbers
Network	Packet / Datagram	Source & destination IP addresses
Data Link	Frame	Source & destination MAC addresses + CRC trailer
Physical	Bits	Encoded signals

1.12 Addressing in Networks

Address Type	Layer	Size / Format	Scope
Physical (MAC) address	Data Link	48 bits, e.g. 07:01:02:01:2C:4B	Local network (hop-to-hop)
Logical (IP) address	Network	32 bits (IPv4) / 128 bits (IPv6)	Universal (end-to-end)
Port address	Transport	16 bits (0-65535)	Identifies a process
Specific (application) address	Application	e.g. email address, URL	Human-friendly identifier

UNIT 2 — Physical and Data Link Layers

PART A — The Physical Layer

2.1 Analog vs Digital

Analog and Digital Data

- **Analog data** — continuous information (human voice, temperature).
- **Digital data** — discrete states (data stored in computer memory: 0s and 1s).

Analog and Digital Signals

- **Analog signal** — infinitely many levels of intensity over time; a smooth curve.
- **Digital signal** — limited number of defined values, usually two (0 and 1); appears as steps.

Periodic and Aperiodic Signals

- **Periodic signal** — completes a pattern within a measurable time frame (a **period**) and repeats it. *Analog periodic signals are the basis of transmission.*
- **Aperiodic (nonperiodic) signal** — no repeating pattern.

Characteristics of a Sine Wave

The sine wave is the most fundamental periodic analog signal, described by three parameters:

1. **Amplitude (A)** — the value of the signal at any point; measured in volts, amperes, or watts. It represents signal strength.
2. **Frequency (f)** — number of periods in one second, measured in **hertz (Hz)**. Frequency = rate of change.
3. $f = 1 / T$ and $T = 1 / f$ (T = period in seconds)
4. High frequency = rapid change; zero frequency = signal never changes (DC).
5. **Phase (φ)** — the position of the waveform relative to time 0, measured in degrees or radians. Describes the shift of the wave.

Wavelength

$\lambda = c / f$ — where c is the propagation speed in the medium and f is frequency. Wavelength is the distance a simple signal travels in one period.

Time Domain vs Frequency Domain

- **Time-domain plot** — shows changes in signal amplitude with respect to time.
- **Frequency-domain plot** — shows the signal's amplitude with respect to frequency. It is more compact: a single sine wave is just one spike.

Composite Signals

A composite signal is made of many simple sine waves. **Fourier analysis** shows that any composite signal can be decomposed into a series of simple sine waves with different frequencies, amplitudes, and phases.

Bandwidth

Bandwidth of a composite signal = the difference between the highest and lowest frequencies it contains.

$$\text{Bandwidth} = f_{\text{highest}} - f_{\text{lowest}}$$

Example: A signal containing frequencies from 1000 Hz to 5000 Hz has a bandwidth of 4000 Hz.

2.2 Digital Signals — Key Terms

- **Bit Rate** — number of bits sent per second (**bps**).
- **Bit Length** — the distance one bit occupies on the medium. Bit length = propagation speed × bit duration.
- **Baud Rate (Signal Rate)** — number of signal elements sent per second.
- Relation: **Bit rate = Baud rate × number of bits per signal element**

Bandwidth Requirement

A digital signal is a composite analog signal with an infinite bandwidth. Since real channels have limited bandwidth, digital transmission always suffers some distortion.

2.3 Transmission Impairments

Signals travel through media and degrade. Three causes:

a) Attenuation

Loss of energy as the signal travels — the signal gets weaker. Compensated using **amplifiers** (analog) or **repeaters** (digital).

Measured in **decibels (dB)**:

$$\text{dB} = 10 \log_{10} (P_2 / P_1)$$

A negative dB means the signal was attenuated; positive means amplified.

b) Distortion

The signal changes its **form or shape**. Happens in composite signals because each frequency component travels at a different propagation speed, so components arrive with different delays — the received signal has a different shape.

c) Noise

Unwanted signals corrupting the transmission: - **Thermal noise** — random motion of electrons in a wire creating extra signal. - **Induced noise** — from motors, appliances acting as sending antennas. - **Crosstalk** — the effect of one wire on another (hearing another conversation). - **Impulse noise** — a spike (a high-energy, short-duration signal) from power lines or lightning.

Signal-to-Noise Ratio (SNR):

$$\text{SNR} = \text{average signal power} / \text{average noise power} \quad \text{SNR(dB)} = 10 \log_{10} \text{SNR}$$

A high SNR means the signal is much stronger than the noise — good quality.

2.4 Data Rate Limits

How fast can we send data over a channel? It depends on bandwidth, signal levels, and channel quality (noise).

Nyquist Bit Rate (Noiseless Channel)

$$\text{BitRate} = 2 \times \text{Bandwidth} \times \log_2(L)$$

Where **L** = number of signal levels. Increasing L increases bit rate but makes the receiver's job harder and increases errors.

Example: Bandwidth = 3000 Hz, L = 2 → BitRate = 2 × 3000 × 1 = 6000 bps.

Shannon Capacity (Noisy Channel)

$$\text{Capacity} = \text{Bandwidth} \times \log_2(1 + \text{SNR})$$

This gives the **theoretical highest** data rate for a noisy channel. Note: the number of signal levels does not appear in the formula.

Example: Bandwidth = 3000 Hz, SNR = 3162 → C = 3000 × log₂(3163) ≈ 34,860 bps.

In practice: Use Shannon to find the upper limit, then Nyquist to find how many signal levels you need.

Performance Measures

- **Throughput** — how fast data actually passes through a point (always ≤ bandwidth).
- **Latency (Delay)** = Propagation time + Transmission time + Queuing time + Processing delay.
- Propagation time = Distance / Propagation speed
- Transmission time = Message size / Bandwidth
- **Bandwidth-Delay Product** = number of bits that can fill the link.
- **Jitter** — variation in delay for different packets.

2.5 Digital Transmission — Line Coding

Line coding converts a sequence of bits into a digital signal.

Line Coding Schemes

1. Unipolar — NRZ - All signal levels are on one side of the time axis (positive or negative only). - Simple but costly in power and has a DC component. Rarely used today.

2. Polar Schemes — voltages on both sides of the axis. - **NRZ-L (Non-Return-to-Zero Level)** — the *level* of the voltage determines the bit value. - **NRZ-I (NRZ Invert)** — a *change* (inversion) in voltage means bit 1; no change means bit 0. Solves part of the synchronization problem. - **RZ (Return to Zero)** — uses three values: positive, negative, zero. The signal returns

to zero in the middle of each bit. Good synchronization but needs more bandwidth. - **Manchester** — combines RZ and NRZ-L ideas. The bit duration is divided into two halves; there is *always* a transition in the middle. Negative→positive = 1, positive→negative = 0 (or vice versa). Self-synchronizing but requires double the bandwidth. - **Differential Manchester** — always a mid-bit transition for synchronization; the presence or absence of a transition at the *start* of the bit determines the bit value. Used in Token Ring.

3. Bipolar Schemes — three levels: positive, zero, negative. - **AMI (Alternate Mark Inversion)** — 0 is represented by zero voltage; 1s alternate between positive and negative. No DC component. - **Pseudoternary** — reverse of AMI (1 = zero voltage).

4. Multilevel & Multitransition Schemes — 2B1Q, 8B6T, 4D-PAM5, MLT-3 — used to increase the bit rate for a given signal rate.

Block Coding

Adds redundancy to ensure synchronization and provide error detection. Notation **mB/nB**: m-bit groups are converted into n-bit groups ($n > m$). - Examples: **4B/5B** (used with NRZ-I in Fast Ethernet), **8B/10B** (Gigabit Ethernet).

Scrambling

Modifies bipolar AMI encoding to avoid long strings of zeros without increasing bandwidth. - **B8ZS (Bipolar with 8-Zero Substitution)** — used in North America. - **HDB3 (High-Density Bipolar 3-zero)** — used in Europe and Japan.

2.6 Analog-to-Digital Conversion

PCM — Pulse Code Modulation

Converts an analog signal into digital data in three steps:

1. **Sampling** — measure the amplitude of the analog signal at equal intervals (PAM — pulse amplitude modulation).
2. **Quantization** — assign each sample an integer value from a finite set of levels.
3. **Encoding** — convert each quantized value into a binary code.

Nyquist Sampling Theorem:

The sampling rate must be **at least twice the highest frequency** contained in the signal. $f_{\text{sampling}} \geq 2 \times f_{\text{max}}$

Example: Human voice occupies up to 4 kHz → sampled at 8000 samples/sec. With 8 bits per sample → 64 kbps (the standard digital voice channel).

Delta Modulation (DM)

Simpler than PCM: instead of encoding the full amplitude, it encodes only the *change* from the previous sample — 1 bit for increase, 0 for decrease.

2.7 Transmission Modes

Parallel Transmission

- n bits sent simultaneously over n separate wires.
- **Advantage:** Speed — n times faster.
- **Disadvantage:** Cost of cabling; limited to short distances (skew problem).

Serial Transmission

One bit sent at a time over a single channel. Cheaper, used for long distances.

a) Asynchronous Serial - Data sent one byte at a time with a **start bit (0)** and one or more **stop bits (1)**. - Gaps between bytes are allowed. - Cheap and effective, but slower. Used for keyboards, modems, RS-232.

b) Synchronous Serial - Bits sent as a continuous stream without start/stop bits. - Timing is derived from the signal itself; requires clock synchronization. - Faster; used in high-speed links.

c) Isochronous — guarantees data arrives at a fixed rate; essential for real-time audio/video.

2.8 Transmission Media

Guided (Wired) Media

1. Twisted-Pair Cable Two insulated copper wires twisted together. Twisting balances out noise/crosstalk. - **UTP (Unshielded Twisted Pair)** — cheap, easy to install, most common in LANs. Categories: Cat 3 (16 MHz), Cat 5/5e (100-125 MHz, 100 Mbps-1 Gbps), Cat 6 (250 MHz), Cat 7 (600 MHz). - **STP (Shielded Twisted Pair)** — has a metal foil shield around each pair; better noise immunity, more expensive and bulkier. - **Connector:** RJ-45. - Applications: Telephone lines, DSL, LANs.

2. Coaxial Cable A central copper conductor surrounded by insulation, a braided metallic shield, and an outer plastic cover. - Carries higher frequency ranges than twisted pair. - **Connector:** BNC. - Applications: Cable TV, older 10Base-2/10Base-5

Ethernet, traditional analog telephone trunks.

3. Fiber-Optic Cable Transmits signals as **light** through a glass or plastic core, using **total internal reflection** (core has a denser medium than the cladding). - **Multimode fiber** — multiple beams travel through the core at different angles. - *Step-index*: abrupt density change at the core-cladding boundary; distortion present. - *Graded-index*: density varies gradually, reducing distortion. - **Single-mode fiber** — very small diameter core; light travels almost horizontally. Minimal distortion, longest distance. - **Connectors**: SC, ST, MT-RJ.

Advantages of Fiber	Disadvantages of Fiber
Huge bandwidth, very high data rate	High installation and maintenance cost
Immune to electromagnetic interference	Unidirectional — two fibers needed for duplex
Very low attenuation → long distances	Fragile; needs expertise to splice
Lightweight, secure (hard to tap)	Specialized equipment required

Unguided (Wireless) Media

Signals are broadcast through free space.

Band	Frequency	Propagation	Applications
Radio waves	3 kHz – 1 GHz	Omnidirectional; sky-wave propagation	AM/FM radio, TV, cordless phones, paging
Microwaves	1 – 300 GHz	Unidirectional; line-of-sight	Cellular phones, satellite, Wi-Fi
Infrared	300 GHz – 400 THz	Short range, cannot pass through walls	TV remotes, IrDA device-to-device

Propagation methods: Ground propagation (low frequency, follows earth's curve), Sky propagation (bounced off the ionosphere), Line-of-sight propagation (very high frequency, straight line).

2.9 Multiplexing

Multiplexing allows the simultaneous transmission of multiple signals over a single data link, to make efficient use of high-bandwidth media. A **MUX** combines streams; a **DEMUX** separates them.

a) FDM — Frequency Division Multiplexing

- **Analog technique.** The bandwidth of the link is divided into frequency bands; each signal is modulated onto a different **carrier frequency**.
- **Guard bands** separate channels to prevent overlap.
- Applications: AM/FM radio, TV broadcasting, first-generation cellular.

b) WDM — Wavelength Division Multiplexing

- Conceptually the same as FDM but applied to **fiber-optic** cable using different wavelengths (colors) of light. A prism or diffraction grating combines/splits the beams.
- **DWDM (Dense WDM)** packs a very large number of channels.

c) TDM — Time Division Multiplexing

- **Digital technique.** The link is shared in **time**, not frequency. Each connection occupies a time slot.
- **Synchronous TDM** — each input gets a fixed, pre-assigned slot whether or not it has data (slots may go empty → wasteful). Needs **framing bits** for synchronization.
- **Statistical (Asynchronous) TDM** — slots are allocated dynamically only to inputs that have data. More efficient; slots must carry an address.

d) CDM / CDMA — Code Division Multiplexing

- All stations transmit at the same time over the whole bandwidth, each using a unique **orthogonal chip code**. The receiver multiplies by the code to recover the desired signal.
- Used in 3G cellular systems.

Spread Spectrum

Used in wireless for security and resistance to jamming: - **FHSS (Frequency Hopping Spread Spectrum)** — the carrier frequency hops among many frequencies in a pseudorandom sequence known to both ends. - **DSSS (Direct Sequence Spread Spectrum)** — each data bit is replaced by a longer chip sequence, spreading the signal over a wider band.

PART B — The Data Link Layer

2.10 Overview and Sublayers

The **Data Link Layer (DLL)** is responsible for **node-to-node (hop-to-hop) delivery**. It takes packets from the network

layer and encapsulates them into **frames**.

It is divided into two sublayers: - **LLC (Logical Link Control)** — IEEE 802.2 — handles flow control, error control, and part of framing; provides a single interface to the network layer. - **MAC (Media Access Control)** — handles framing and, most importantly, decides *who* can use the shared medium (covered in Unit 3).

Functions of the Data Link Layer

1. **Framing** — packaging bits into frames.
2. **Physical addressing** — adding MAC addresses.
3. **Flow control** — pacing the sender to the receiver's capability.
4. **Error control** — detecting and correcting/retransmitting damaged frames.
5. **Access control** — determining which device controls the link (MAC sublayer).

2.11 Framing

Framing divides the continuous bit stream into distinguishable units.

Fixed-Size Framing

No need for boundaries; size itself acts as a delimiter. *Example: ATM cells (53 bytes).*

Variable-Size Framing

Needs a way to mark the end of one frame and start of the next.

1. Character-Oriented (Byte-Oriented) Framing - Data is a sequence of 8-bit characters; a **flag** (a special byte) marks the beginning and end. - **Problem:** If the flag pattern appears inside the data, the receiver thinks the frame ended. - **Solution — Byte Stuffing:** insert an extra **escape character (ESC)** before any byte that looks like a flag or an ESC. The receiver removes the ESC before delivering.

2. Bit-Oriented Framing - Data is a sequence of bits; the flag is the bit pattern **01111110**. - **Solution — Bit Stuffing:** the sender inserts a **0** after five consecutive 1s in the data. The receiver removes any 0 that follows five consecutive 1s. - Used by HDLC, PPP.

2.12 Error Detection and Correction

Types of Errors

- **Single-bit error** — only one bit in a data unit changes (0→1 or 1→0). Common in parallel transmission.
- **Burst error** — two or more bits in the data unit change. Length = distance from the first corrupted bit to the last. More common in serial transmission because noise usually lasts longer than one bit duration.

Redundancy

The central concept: send **extra (redundant) bits** along with the data so the receiver can check consistency.

Correction Approaches

- **Forward Error Correction (FEC)** — the receiver detects *and corrects* the error itself using enough redundancy.
- **Retransmission (ARQ)** — the receiver detects the error and asks the sender to resend.

a) Parity Check (VRC)

Simple Parity Check: One redundant bit (parity bit) is appended so the total number of 1s becomes even (even parity) or odd (odd parity). - Detects **all odd-numbered** bit errors. - Cannot detect an even number of errors. Weakest method.

Two-Dimensional Parity Check (LRC): Data is organized in a table; a parity bit is computed for each row *and* each column. Much stronger — detects most burst errors up to the row length.

b) Cyclic Redundancy Check (CRC)

The most powerful and widely used detection method, based on **binary (modulo-2) division**.

Process: 1. A string of n zeros is appended to the data unit (n = degree of the generator polynomial, i.e., one less than the number of bits in the divisor). 2. The extended data is divided by a predetermined **generator (divisor)** using modulo-2 division (XOR). 3. The **remainder** (the CRC, n bits) replaces the appended zeros. 4. The receiver divides the received unit by the same divisor. **Remainder = 0 → frame accepted**; otherwise discarded.

Common Generator Polynomials: | Name | Polynomial | |---|---| | CRC-8 | $x^8 + x^2 + x + 1$ | | CRC-10 | $x^{10} + x^9 + x^5 + x^4 + x^2 + 1$ | | CRC-16 | $x^{16} + x^{12} + x^5 + 1$ | | CRC-32 | $x^{32} + x^{26} + x^{23} + \dots + x + 1$ (used in Ethernet) |

Strengths: CRC can detect all single-bit errors, all double-bit errors (with a suitable polynomial), any odd number of errors, and all burst errors shorter than the degree of the polynomial.

Used at the network and transport layers (IP, TCP, UDP).

Sender: 1. Divide data into k sections of n bits each. 2. Add all sections using **one's complement arithmetic** (carry wrapped around). 3. Complement the sum → this is the **checksum**. 4. Send data + checksum.

Receiver: Adds all sections including the checksum. If the result (complemented) is **all 0s**, the data is accepted; otherwise rejected.

Checksum is simpler than CRC but weaker.

d) Hamming Code (Error Correction)

A **forward error correcting** code that can correct single-bit errors.

- Redundant bits are placed at positions that are powers of 2 (1, 2, 4, 8, ...).
- Number of redundancy bits r for m data bits must satisfy: $2^r \geq m + r + 1$
- Each parity bit checks a specific set of bit positions.
- At the receiver, recomputing the parity bits yields a binary number (**syndrome**) that points directly to the position of the erroneous bit, which is then flipped.

Hamming Distance: the number of differing bits between two codewords, found by XOR-ing them and counting the 1s. - To detect d errors: minimum Hamming distance must be $d + 1$. - To correct d errors: minimum Hamming distance must be $2d + 1$.

2.13 Flow Control and Error Control Protocols

Flow control — a set of procedures to restrict how much data the sender can transmit before waiting for acknowledgement.

Error control — detection and retransmission of damaged/lost frames, usually implemented as **ARQ (Automatic Repeat Request)**.

Protocols for Noiseless Channels

1. Simplest Protocol - No flow or error control. Assumes the receiver can instantly handle any incoming frame. Purely theoretical.

2. Stop-and-Wait Protocol - The sender sends one frame, then **stops and waits** for an acknowledgement (ACK) before sending the next. - Provides flow control but no error control. - Very inefficient over long-delay links.

Protocols for Noisy Channels

3. Stop-and-Wait ARQ - Adds error control to Stop-and-Wait: sequence numbers (alternating 0 and 1), ACKs, a **timer**, and retransmission on timeout. - The sender keeps a copy of the last frame until its ACK arrives. - **Efficiency:** poor on high bandwidth-delay links. - Sender window size = 1, Receiver window size = 1.

4. Go-Back-N ARQ - The sender can send up to **N** frames (window size) before needing an ACK — **pipelining**. - If a frame is lost or damaged, the sender **retransmits that frame and all frames after it**. - Receiver window size = **1** (accepts frames only in order; discards out-of-order frames). - Sender window size $\leq 2^m - 1$ (where m = number of bits for sequence numbers). - Simple receiver, but wasteful when errors occur.

5. Selective Repeat ARQ - Only the **damaged or lost frame** is retransmitted. - Receiver window size = sender window size = $2^{(m-1)}$. - The receiver buffers out-of-order frames and reorders them. - More efficient use of bandwidth, but requires more complex logic and buffering at the receiver, plus **negative acknowledgements (NAKs)**.

Comparison

Feature	Stop-and-Wait ARQ	Go-Back-N ARQ	Selective Repeat ARQ
Sender window	1	$2^m - 1$	$2^{(m-1)}$
Receiver window	1	1	$2^{(m-1)}$
Retransmission	The one frame	Frame + all following	Only the bad frame
Receiver buffering	None	None	Required
Complexity	Lowest	Moderate	Highest
Bandwidth efficiency	Lowest	Moderate	Highest

Piggybacking

To improve efficiency in bidirectional communication, the **acknowledgement is attached to an outgoing data frame** rather than sent separately. This saves bandwidth.

Sliding Window

A conceptual mechanism where both sender and receiver maintain a "window" of sequence numbers they are allowed to send/accept. The window slides forward as frames are acknowledged.

2.14 Data Link Layer Protocols

HDLC — High-Level Data Link Control

A bit-oriented protocol for point-to-point and multipoint links.

Transfer modes: - **NRM (Normal Response Mode)** — an unbalanced configuration: one primary station, one or more secondaries. Secondaries respond only when polled. - **ABM (Asynchronous Balanced Mode)** — a balanced configuration: both stations are peers and either can send. This is the common mode today.

Frame format: Flag | Address | Control | Information | FCS | Flag - **Flag:** 01111110 - **FCS:** Frame Check Sequence (CRC-16 or CRC-32)

Frame types: - **I-frames (Information)** — carry user data and control information (piggybacked ACK). - **S-frames (Supervisory)** — carry only control info: RR (Receive Ready), RNR (Receive Not Ready), REJ (Reject), SREJ (Selective Reject). - **U-frames (Unnumbered)** — link management: connection setup, disconnection, mode setting.

PPP — Point-to-Point Protocol

The most common protocol for point-to-point links (dial-up, DSL, leased lines). Byte-oriented.

Frame format: Flag | Address | Control | Protocol | Payload | FCS | Flag

Services provided: - Framing and error detection - Multiple network layer protocol support (via the Protocol field) - Authentication — **PAP** (Password Authentication Protocol, two-way handshake, simple) and **CHAP** (Challenge Handshake Authentication Protocol, three-way handshake, more secure) - Network address configuration

Component protocols: - **LCP (Link Control Protocol)** — establishing, maintaining, configuring, and terminating links. - **NCP (Network Control Protocol)** — a set of protocols to configure different network layer protocols; **IPCP** configures IP. - **AP (Authentication Protocols)** — PAP and CHAP.

PPP Transition Phases: Dead → Establish → Authenticate → Network → Open → Terminate.